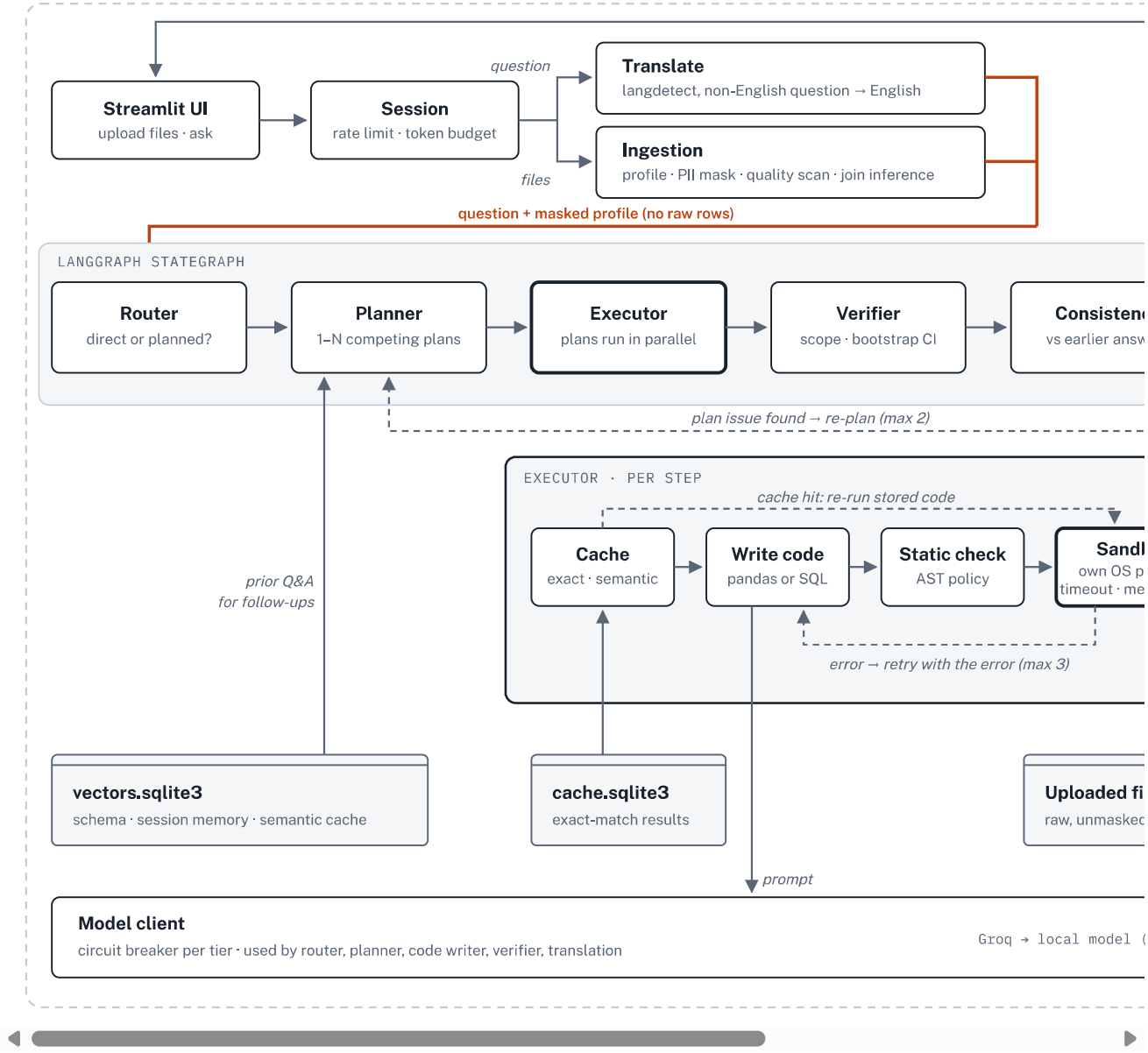
CAPSTONE · SYSTEM ARCHITECTURE

Autonomous Data-Analysis Agent

How one question travels from upload to a verified answer. The agent plans, writes code, runs it in a local sandbox, checks the result, and compares it with earlier answers.



The orange path is the only data that leaves your machine: the question and a PII-masked profile of each column. The real rows are read only inside the sandbox process, which runs locally. Dashed arrows are the agent's loops: a cache shortcut, a retry when code fails, and a re-plan when a result shows the plan was wrong.

— Privacy boundary

Ingestion masks names, emails, phone numbers and ID numbers before any model sees the profile.

--- Self-correction

A failed step retries up to three times with the error message added. A step that exposes a wrong

Calculations run on the original data, so answers stay exact.

Around the loop

A thumbs-down triggers an independent re-derivation in a fresh session. A thumbs-up adds the assumption sends the planner back to revise the remaining steps, up to twice.